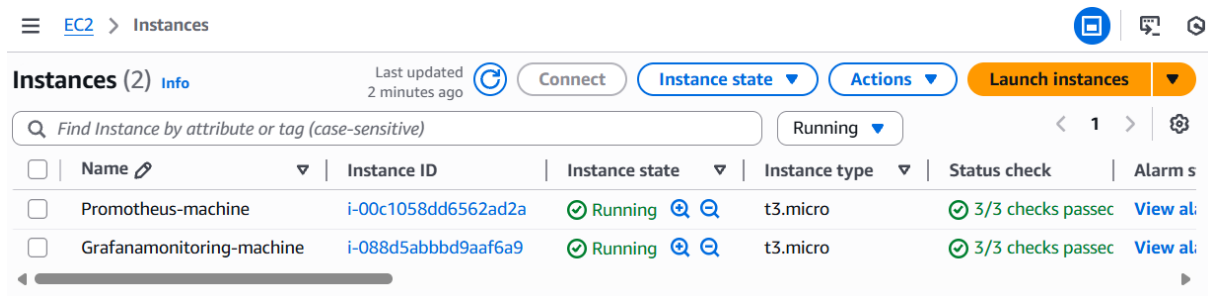


Prometheus & Grafana Practical

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1. Docker Monitoring using Prometheus and Grafana

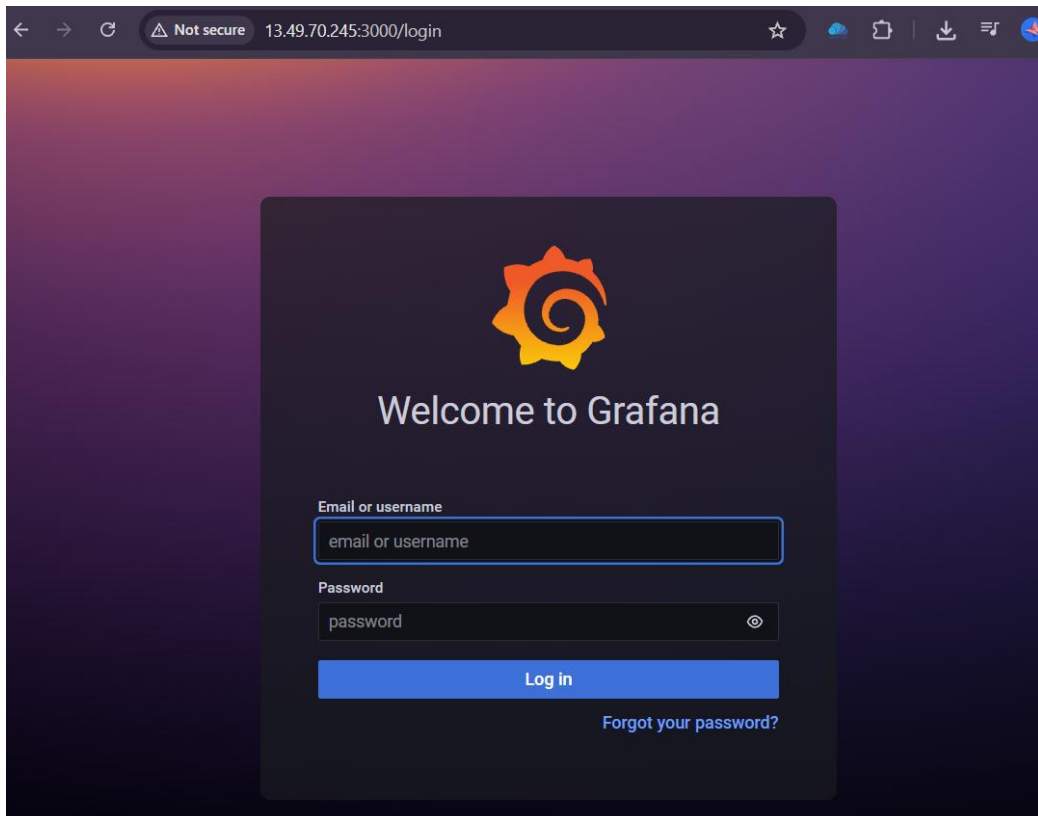
- Create two Ubuntu EC2 machines one for Prometheus and one for Grafana.
- Installation plan:
 - Grafana machine: Install Grafana.
 - Prometheus machine: Install Docker and Prometheus.



The screenshot shows the AWS Management Console 'Instances' page. It displays two EC2 instances: 'Prometheus-machine' and 'Grafana-monitoring-machine'. Both instances are in a 'Running' state, using 't3.micro' instance types. The status checks for both instances show '3/3 checks passed'. The page includes a search bar, filters, and a table with columns for Name, Instance ID, Instance state, Instance type, Status check, and Alarm s.

Name	Instance ID	Instance state	Instance type	Status check	Alarm s
Prometheus-machine	i-00c1058dd6562ad2a	Running	t3.micro	3/3 checks passed	View al
Grafana-monitoring-machine	i-088d5abbbd9aaf6a9	Running	t3.micro	3/3 checks passed	View al

- Grafana-machine setup:
 - Grafana server runs on **port 3000**, so enable this port in the security group.
 - Access the Grafana webpage using the **public IP** of the machine and log in



- Prometheus-machine setup:
 - Install the Docker and Prometheus in the machine

```
root@ip-172-31-0-161:/home/ubuntu# ls
prometheus-2.34.0.linux-amd64  prometheus-2.34.0.linux-amd64.tar.gz
root@ip-172-31-0-161:/home/ubuntu# cd prometheus-2.34.0.linux-amd64
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64#
```

- Configure Docker to allow Prometheus to track it via **port 9323** by adding the required data to `/etc/docker/daemon.json`, then restart Docker.

```
{
    "metrics-addr" : "0.0.0.0:9323",
    "experimental" : true
}

~
~
~
```

- Enable ports 9323 and 9090 in the security group of the Prometheus machine.

- To see Docker stats, access the Prometheus machine using **public IP:9323**.

```

← → ↻ ⚠ Not secure 16.170.217.37:9323/metrics ☆ ☁ 📦 ⬇

# HELP builder_builds_failed_total Number of failed image builds
# TYPE builder_builds_failed_total counter
builder_builds_failed_total{reason="build_canceled"} 0
builder_builds_failed_total{reason="build_target_not_reachable_error"} 0
builder_builds_failed_total{reason="command_not_supported_error"} 0
builder_builds_failed_total{reason="dockerfile_empty_error"} 0
builder_builds_failed_total{reason="dockerfile_syntax_error"} 0
builder_builds_failed_total{reason="error_processing_commands_error"} 0
builder_builds_failed_total{reason="missing_onbuild_arguments_error"} 0
builder_builds_failed_total{reason="unknown_instruction_error"} 0
# HELP builder_builds_triggered_total Number of triggered image builds
# TYPE builder_builds_triggered_total counter
builder_builds_triggered_total 0
# HELP engine_daemon_container_actions_seconds The number of seconds it takes to process each container action
# TYPE engine_daemon_container_actions_seconds histogram
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.005"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.01"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.025"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.05"} 1
engine_daemon_container_actions_seconds_bucket{action="changes",le="0.1"} 1

```

- Update prometheus.yml by adding - *job_name: "docker"* to scrape Docker metrics.

```

root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# vi prometheus.yml
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64#

```

```

- targets:
  # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
  - job_name: "docker"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

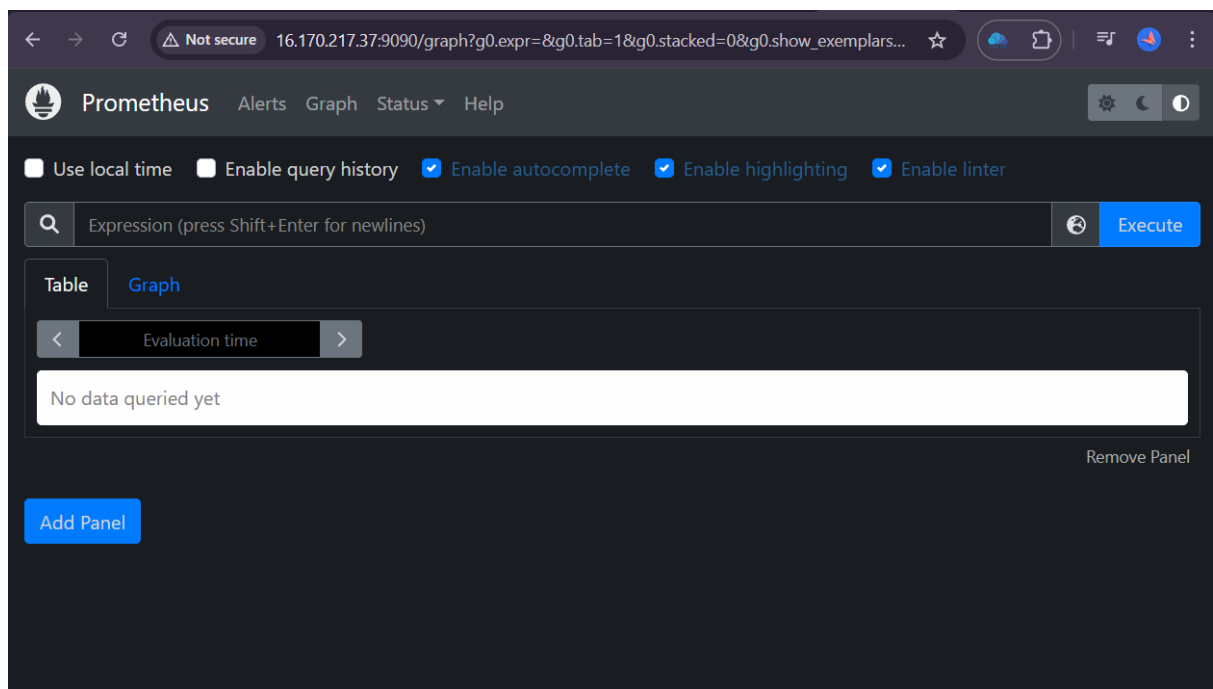
    static_configs:
      - targets: ["localhost:9323"]

```

- Start Prometheus by running `./prometheus`

```
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# vi prometheus.yml
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ./prometheus
ts=2026-02-12T09:49:17.164Z caller=main.go:479 level=info msg="No time or size retention was set so using the default time retention" duration=15d
ts=2026-02-12T09:49:17.164Z caller=main.go:516 level=info msg="Starting Prometheus" version="(version=2.34.0, branch=HEAD, revision=881111fec4332c33094a6fb2680c71fffc427275)"
ts=2026-02-12T09:49:17.164Z caller=main.go:521 level=info build_context="(go=go1.17.8, user=root@121ad7ea5487, date=20220315-15:18:00)"
ts=2026-02-12T09:49:17.164Z caller=main.go:522 level=info host_details="(Linux 6.14.0-1018-aws #18~24.04.1-Ubuntu SMP Mon Nov 24 19:46:27 UTC 2025 x86_64 ip-172-31-0-161 (none))"
ts=2026-02-12T09:49:17.164Z caller=main.go:523 level=info fd_limits="(soft=1024, hard=1048576)"
ts=2026-02-12T09:49:17.164Z caller=main.go:524 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-02-12T09:49:17.168Z caller=web.go:540 level=info component=web msg="Start listening for connections" address=0.0.0.0:9090
ts=2026-02-12T09:49:17.168Z caller=main.go:937 level=info msg="Starting TSDB ..."
```

- Access the Prometheus webpage using public IP:9090



← → ↻ Not secure 16.170.217.37:9090/targets ☆ [cloud] [share] [list] [refresh]

Prometheus Alerts Graph Status ▾ Help [settings] [moon] [bell]

Targets

All Unhealthy Collapse All 🔍 Filter by endpoint or labels

docker (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9323/metrics	UP	<code>instance="localhost:9323"</code> <code>job="docker"</code>	2.622s ago	2.609ms	

prometheus (1/1 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	<code>instance="localhost:9090"</code> <code>job="prometheus"</code>	10.683s ago	3.892ms	

- Connect Grafana to Prometheus:
 - In Grafana, add Prometheus as a data source so that Grafana can visualize the metrics collected by Prometheus.

← → ↻ Not secure 13.49.70.245:3000/datasources/new ☆ [cloud] [share] [list] [refresh]

Add data source
Choose a data source type

🔍 Filter by name or type [← Cancel](#)

Time series databases

Prometheus
Open source time series database & alerting
[Core](#)

Graphite
Open source time series database
[Core](#)

OpenTSDB
Open source time series database
[Core](#)

InfluxDB
Open source time series database
[Core](#)

← → ↻ ⚠ Not secure 13.49.70.245:3000/datasources/edit/sqr-cFwk



Data Sources / Prometheus

Type: Prometheus

⚙ Settings

📊 Dashboards

Name ⓘ Prometheus

Default ☒

HTTP

Public IP of Prometheus WebPage

URL ⓘ http://16.170.217.37:9090/

Access Server (default)

Help >

Allowed cookies ⓘ New tag (enter key to add)

Timeout ⓘ Timeout in seconds

Auth

Basic auth ☐ With Credentials ⓘ ☐

TLS Client Auth ☐ With CA Cert ⓘ ☐

Skip TLS Verify ☐

Forward OAuth Identity ⓘ ☐

Alertmanager data source ⓘ Choose

Scrape interval ⓘ 15s

Query timeout ⓘ 60s

HTTP Method ⓘ POST ▾

Misc

Disable metrics lookup ⓘ ☐

Custom query parameters ⓘ Example: max_source_resolution=5m&timeout=10

Exemplars

+ Add

✓

Data source is working

Back

Explore

Delete

Save & test

